

Towards Inclusive University Campuses: Optimal Planning of Wheelchair Charging Infrastructure

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Abstract—This paper proposes an optimization-based framework for the strategic deployment of wheelchair charging stations on the Kuwait College of Science and Technology (KCST) campus. The approach employs a greedy maximal-coverage algorithm to identify optimal charger placements that maximize accessibility while minimizing infrastructure costs and redundancy. Unlike exhaustive combinatorial search, the greedy method achieves near-optimal results with significantly reduced computational complexity, making it practical for real-world facility planning. MATLAB-based simulations are performed accounting for demand distribution and designated accessibility hotspots. Results show that about 95% coverage can be achieved with only a few strategically placed chargers, while dispersed layouts require more installations with diminishing marginal returns. The presented redundancy analysis further demonstrates that controlled overlaps can improve resilience without excessive infrastructure. The proposed framework thus provides a lightweight, reproducible, and scalable method for inclusive energy access planning, with applicability to universities, airports, malls, and other large facilities.

Keywords—Wheelchair Charging Stations, Accessibility planning, Greedy Optimization, Demand Points, Coverage, Redundancy

I. INTRODUCTION AND LITERATURE REVIEW

Ensuring equitable access to campus resources increasingly requires thinking beyond ramps and signage to the everyday energy needs of mobility devices. Globally, about 15% of the population—roughly 1.3 billion people—live with disabilities, facing persistent barriers in education, work, and mobility [1–3]. Locally at Kuwait College of Science and Technology (KCST), only 18 out of around 2000 students (0.9%) currently have physical disabilities, yet this small percentage underscores the need for accessible infrastructure rather than diminishing its importance. The lack of facilities, such as wheelchair charging stations, may discourage enrollment of students with special needs.

Recent evidence further underscores that the built environment itself is a decisive factor shaping participation for people with long-term physical disabilities. Even when assistive technologies such as power wheelchairs are available, inaccessible infrastructure continues to restrict independence and day-to-day engagement [4]. This highlights the importance of coupling mobility devices with supportive infrastructure, such as strategically planned charging facilities, to move beyond minimum compliance and foster genuine inclusion.

For wheelchair users in particular, energy access is a foundational but often overlooked dimension of accessibility. Power wheelchairs rely on regular charging cycles; a missed

charge can strand a user, disrupt classes or work, and in some cases pose safety risks.

A common misconception is that power wheelchairs can be recharged using any standard wall socket. In reality, these devices require specialized chargers that regulate voltage and current to protect the batteries and ensure safety. While the charging unit itself is plugged into a regular socket, the availability of properly sited, dedicated charging stations is what makes energy access practical in real-world settings. Without designated locations, users may face inaccessible outlets, unsafe cable routing, or a lack of space to maneuver, making the presence of standard sockets alone insufficient to meet accessibility needs. Airports and other public facilities have begun deploying accessible charging infrastructure for mobility devices. For instance, Miami International Airport (MIA) became the first airport in the U.S. to install wheelchair charging stations airport-wide [5], as shown in Fig. 1. These stations allow passengers with electric wheelchairs and mobility scooters to recharge while waiting for flights, reducing travel stress and increasing accessibility. Ten units were strategically distributed across multiple concourses and international arrivals areas, ensuring coverage and convenience. This example highlights the growing recognition of charging access as a core component of inclusive design.

Practical advice to users typically focuses on *how* to charge (e.g., recommended connectors, charge times, and battery care) rather than *where* and *how many* chargers should be placed across complex buildings [6]. At the same time, organizations and manufacturers have begun to publish best-practice guidance for barrier-free charging infrastructure—highlighting clear approaches such as generous maneuvering space, reach-range compliant sockets, and intuitive, well-marked locations near natural gathering points—yet these guidelines rarely answer the campus-scale planning problem [7].

Qualitative studies document how power-wheelchair users navigate buildings, emphasizing the importance of predictable wayfinding, redundancy, and access to reliable charging opportunities as part of “daily life logistics” [8]. These insights argue for charger placement that follows usage patterns (entrances, hubs, and dwell areas) rather than purely geometric rules. Moreover, a stream of engineering papers focuses on the charger or wheelchair hardware.



Fig. 1. Wheelchair charging station installed at Miami International Airport (MIA) [6]

For example, solar-assist concepts were discussed to extend range and reduce grid dependency, in addition to compact DC charging modules for mobility devices. While valuable, these solutions typically target how to supply energy rather than where to situate outlets within real facilities [9]. Industry and disability-advocacy resources, codify minimum clearances, line-of-travel considerations, signage, cable management, and height/force requirements for sockets and interfaces. These documents provide the *micro-design* criteria needed once a location is chosen, but they do not solve the macro-level allocation of “how many chargers per building wing/floor” or the trade-offs among coverage, redundancy, and cost [10].

Despite growing attention to inclusive infrastructure, there remains a lack of campus-scale, data-driven methods that decide about : 1) the number of wheelchairs charging points needed to meet coverage goals; 2) their optimal locations given travel radii and building topology; and 3) the redundancy levels to hedge against occupancy, outages, or maintenance

What is missing is a simple, reproducible planning workflow that starts from real architectural drawings, incorporates priority “hotspots” (e.g., entrances, auditoria, labs, admin counters), and yields transparent, multi-objective trade-offs between coverage, redundancy, and infrastructure effort.

In this paper, a simulation-based map-driven optimization framework is proposed to guide the placement of wheelchair charging stations on the KCST campus. The proposed technique models demand points on each floor, incorporates stakeholder-defined priority hotspots, and employs a greedy maximal-coverage heuristic to allocate a given number of chargers K . For each iteration, the framework reports coverage levels, redundancy (points within range of at least two chargers), and recommends the smallest K that achieves diminishing marginal returns (“elbow point”). Designed to be lightweight, transparent, and reproducible, the approach is accessible to facilities teams and accessibility committees without requiring advanced optimization expertise. Beyond addressing KCST’s immediate needs, the framework is readily adaptable to larger facilities such as malls, stadiums, and universities, aligning with WHO and UN accessibility principles by emphasizing functionally reachable energy access rather than nominal outlet counts.

By addressing the gap, the campus can foster inclusivity, attract more diverse students in the future, and align with international standards for sustainable and accessible university design. This effort resonates with global policy

frameworks, particularly the United Nations Sustainable Development Goals (SDGs). Accessible charging infrastructure for mobility devices directly supports SDG 4 (Quality Education) by enabling equal participation of students with disabilities in academic life, SDG 10 (Reduced Inequalities) by addressing systemic barriers to mobility and independence, and SDG 11 (Sustainable Cities and Communities) by promoting inclusive and resilient campus design. Embedding these principles into planning ensures that accessibility initiatives are not only locally impactful but also aligned with international commitments to equitable and sustainable development [11].

II. SYSTEM DESCRIPTION AND MATHEMATICAL MODELING

KCST campus comprises nine interconnected academic blocks. The blocks are functionally differentiated to concentrate footfall and dwell time around teaching, laboratory, administrative, and student-life activities—precisely the locations where wheelchair users are most likely to require convenient, reliable access to charging. Table I summarizes the program per block and per floor; these functions informed the candidate demand points, and hotspot selections used in the simulations. Fig. 2 shows a visual representation of the campus blocks for each floor.

Large lecture halls, the cafeteria, administrative counters, and the library were treated as priority zones, as these are the spots where students spend most of their time at. The surrounding outdoor area of the ground floor was taken into account as it represents the parking slots, sports facilities, and spots where seasonal events take place.

The charger placement is formulated as a weighted maximal-coverage problem over a finite set of demand locations extracted from the floor layouts. The demand points are the points where the special needs students are at frequently. Demand points can be lecture halls, cafeteria, staff offices, etc..

Let $V = \{1, 2, \dots, N\}$ be the set of demand points on a floor, where N is the number of the demand points. A charger with optimal allocation can be placed at any demand point $j \in V$, where j is demand point chosen to place the charger at.

For a charger at point j , we define a coverage neighborhood with radius R , where the radius is defined as the maximum distance that wheelchair users can reasonably travel from their main activity points before needing access to a charger.



(a)

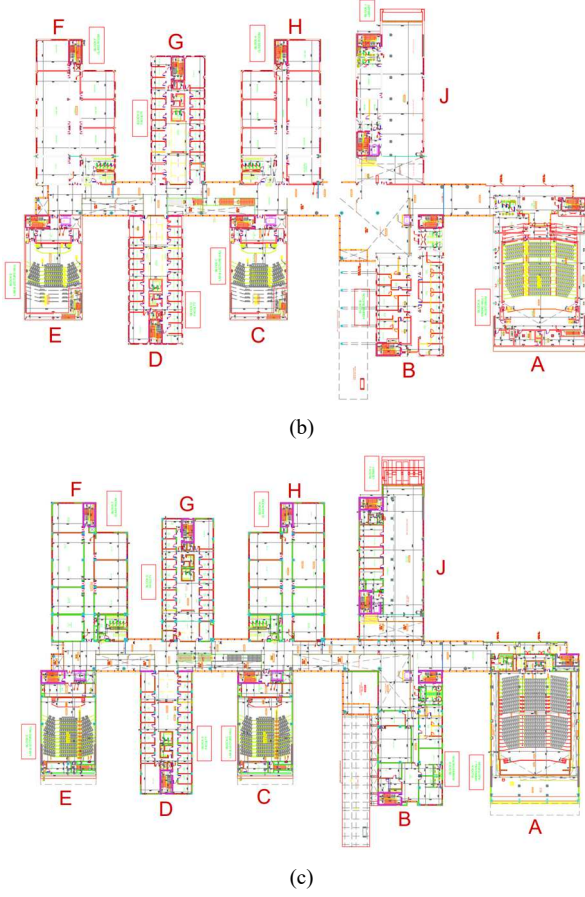


Fig. 2. KCST campus layout for all floors: (a) ground floor (b) first floor (c) second floor

In this study, the charger coverage radius was set to 50 meters, reflecting a practical balance between accessibility and infrastructure efficiency. A 50 m walking or wheeling distance is generally considered reasonable for indoor mobility, particularly in academic and public buildings, where facilities such as lecture halls, laboratories, and offices are typically spaced within this range.

The coverage neighborhood of point j is defined as the set of demand points i that fall within distance R of charger j

$$N_R(j) = \{i \in V : d(i, j) \leq R\} \quad (1)$$

Where $d(i, j)$ is the walking distance between demand point i and candidate charger j .

For regular demand points, we assign a priority weight of 1 ($w_i = 1$), whereas hotspot demand points receive a weight of 3 ($w_i = 3$). Given a number of K chargers, the goal is place the chargers so that they cover as many demand points as possible. Mathematically, the objective function is represented by

$$\max_{S \subseteq V, |S| \leq K} f(S) = \sum_{i \in V} w_i \cdot 1(\exists j \in S : i \in N_R(j)) \quad (2)$$

Where $S \subseteq V$ represents the selected charger locations, $|S| \leq K$ is the budget constraint, and the indicator function $1(\cdot)$ evaluates to 1 if the demand point i is covered by at least one charger, and 0 otherwise.

TABLE I CAMPUS DETAILS BY BLOCK AND FLOOR

Block	Ground Floor (G)	First Floor (F1)	Second Floor (F2)
A	Dining room (cafeteria)	Auditorium	
B	Administration, Marketing, Facilities, Student Affairs	IT, Finance, and HR Departments	College Management
C	Laboratories	Large Lecture Hall (spans F1–F2)	
D	Offices for staff & academics		
E	Laboratories	Large Lecture Hall (spans F1–F2)	
F	Laboratories	Lecture Halls	
G	Offices for staff & academics		
H	Laboratories	Lecture Halls	
J	Research & fabrication labs; teaching studios	Library	

In words, the objective is to maximize the total weighted number of demand points that lie within reach of at least one charger [12].

In addition to coverage, we introduce a redundancy metric to account for resilience and user experience. Redundancy reflects the fraction of demand points that are simultaneously covered by two or more chargers, thereby reducing the risk of service disruption or excessive queuing. It is defined as

$$\rho(s) = \frac{1}{|V|} \sum_{i \in V} 1(|\{j \in S : i \in N_R(j)\}| \geq 2) \quad (3)$$

Where $\rho(s)$ is the fraction nodes covered by at least two chargers [13]. Eq. (3) computes the fraction of demand points covered by two or more chargers, reflecting redundancy.

The charger placement problem is a variant of the maximum coverage problem. This type of problems is well-known to be nondeterministic polynomial time hard (NP-hard) problem, which means that no known algorithm can guarantee finding the exact optimal solution efficiently when the problem size grows. However, the objective function $f(\cdot)$ possesses two favorable properties. The first is monotonicity which indicates that adding an additional charger can never reduce the overall coverage. The second is submodularity where the principle of diminishing returns applies; while the first few chargers yield large improvements in coverage, later chargers contribute progressively less [14]. Therefore, the use of greedy algorithm is favorable as it provides a guaranteed approximation ratio of $1 - 1/e \approx 63\%$ of the optimal solution. In practice, especially in structured environments such as building layouts, greedy selection often performs much closer to optimal [15].

Greedy builds S iteratively, starting from $S_0 = \emptyset$, at step $t = 0, 1, \dots, K - 1$ choose,

$$j^* \in \arg \max_{j \in V \setminus S_t} \Delta(j | S_t) \quad (4)$$

Where j^* is the new site that gives the largest marginal gain at this iteration, S_t is the set of chargers sites after t steps, $V \setminus S_t$ is the all unused candidate sites. Then, it sets $S_{t+1} = S_t \cup \{j^*\}$.

The marginal gain equation $\Delta(j | S_t)$ has a closed form,

$$\Delta(j | S_t) = \sum_{i \in V} w_i \cdot 1(i \in N_R(j) \wedge i \notin \bigcup_{l \in S_t} N_R(l)) \quad (5)$$

Eq. 5 quantifies how much new weighted coverage is added if a charger is placed at candidate j , given the current set of installed chargers S . The indicator logic function is true if two conditions hold simultaneously. First, the demand point i lies within radius R of site j . The second condition is that the demand point is not already covered by any charger in the current set S , that is, it does not belong to the union of neighborhoods $\bigcup_{l \in S} N_R(l)$. In practice, this means the sum explicitly counts the weighted number of demand points that become newly covered by adding site j .

Other optimization methods could also be used for charger placement, but each comes with trade-offs. For instance, the Exact Integer Programming approach guarantees the global optimum by formulating the problem with binary decision variables. While powerful, it is computationally expensive: the runtime grows very quickly with the number of candidate sites, chargers, or radius scenarios. It can also require careful solver tuning and is harder to explain to non-specialist stakeholders. In practice, this method is only feasible for very small problems and is not practical for multi-floor campus-scale analysis. Another example is the Local Search method, where it starts with a solution and tries to improve it by swapping out one chosen site with another unchosen site if that increases coverage. It can sometimes improve on greedy, but it does not provide a worst-case guarantee and can get stuck in “plateaus”. The extra computation and coding effort usually yield only small improvements compared to greedy.

Therefore, the greedy algorithm offers the best balance, as it is simple, explainable, fast enough for real floor plans, and comes with a theoretical guarantee close to optimal, unlike most alternatives.

III. SIMULATION RESULTS

The proposed greedy algorithm was implemented and validated using MATLAB. To clarify the sequence of operations, Fig. 3 presents the flowchart of the proposed methodology. The simulation starts with loading the floor layout and scale calibration. The calibration process is carried out by choosing two points on the layout with known distance. The system then computes a calibration factor (px2m), converting pixel distances into meters. For any two demand points i and j , the Euclidian walking distance is calculated as:

$$d(i, j) = px2m \cdot \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (6)$$

Where (x_i, y_i) and (x_j, y_j) are the pixel coordinates. This step ensures that all subsequent coverage analyses are carried out in physically meaningful units (meters).

Next, the used identifies all potential demand points on each floor layout. Each demand point is stored in the candidate set V defined previously. This step effectively digitize the functional usage of the campus, linking charger placement to actual student and staff movement patterns.

Then, the floor map reopens and the Hotspot points are selected, assigning a weight of 3 to them ($w_i = 3$), whereas regular demand points have weight of 1 ($w_i = 1$). The weighting scheme biases the optimization toward covering locations with the highest functional importance, reflecting the principle of universal design where accessibility is prioritized in key social and academic spaces.

For each floor, and for a budget of $K \in \{1, 2, 3, 4, 5\}$ chargers, the algorithm selects charger locations using the weighted greedy maximal-coverage approach. At each step, the candidate point that maximize the weighted marginal gain is chosen, ensuring that every additional charger provides the greatest possible improvement in coverage. Finally, the algorithm computes marginal gains as K increases, identifying the smallest number of chargers K^* at which the incremental coverage improvement drops below a threshold of 3%. This “elbow rule” provides a natural recommendation for charger count. Additionally, sensitivity analyses are performed across different coverage radii ($R = 40, 50, 60$ m), allowing stakeholders to examine how assumptions about wheelchair range, cord length, or walking tolerances influence placement outcomes.

Fig. 4. shows the iterations of placing chrgers with displaying the coverage and redundancy percentages for the ground floor. Starting with $K=1$, the coverage percentage is 56.6%, and redundancy equals to zero since only one charger exists. Incrementing K to be 2, the coverage increases to 75.5%, and the redundancy becomes 26.4%. Further, as K is incrementd to 3, the coverage increases to 92.5%, with redundancy of 43.4%. With $K=4$, the coverage is 98.1% and redundancy of 60.4%. Therefore, with $K=5$ and coverage of 100%, the marginal gain would be less than 3%, making $K=5$ a recommended choice for the ground floor layout.

While the greedy algorithm was adopted as the main optimization tool due to its scalability and near-optimal guarantees, its performance was further validated against alternative placement strategies. In particular, exhaustive search (which evaluates all possible site combinations) and random allocation (averaged across multiple trials) were implemented as baselines.

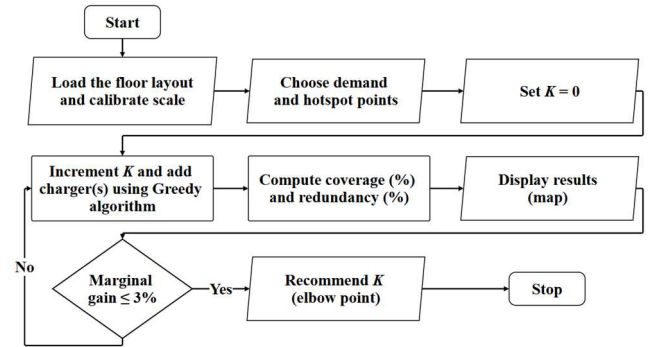
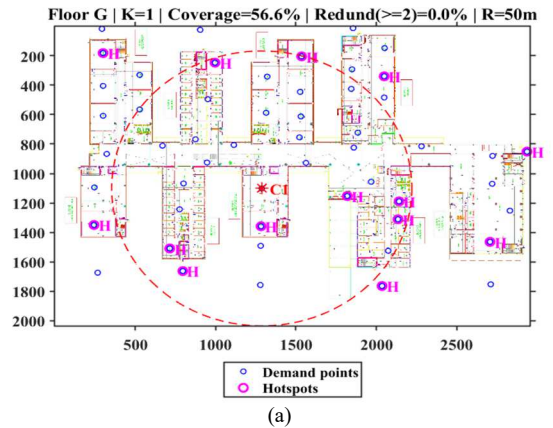


Fig. 3. Flowchart of the proposed methodology for charger placement optimization.



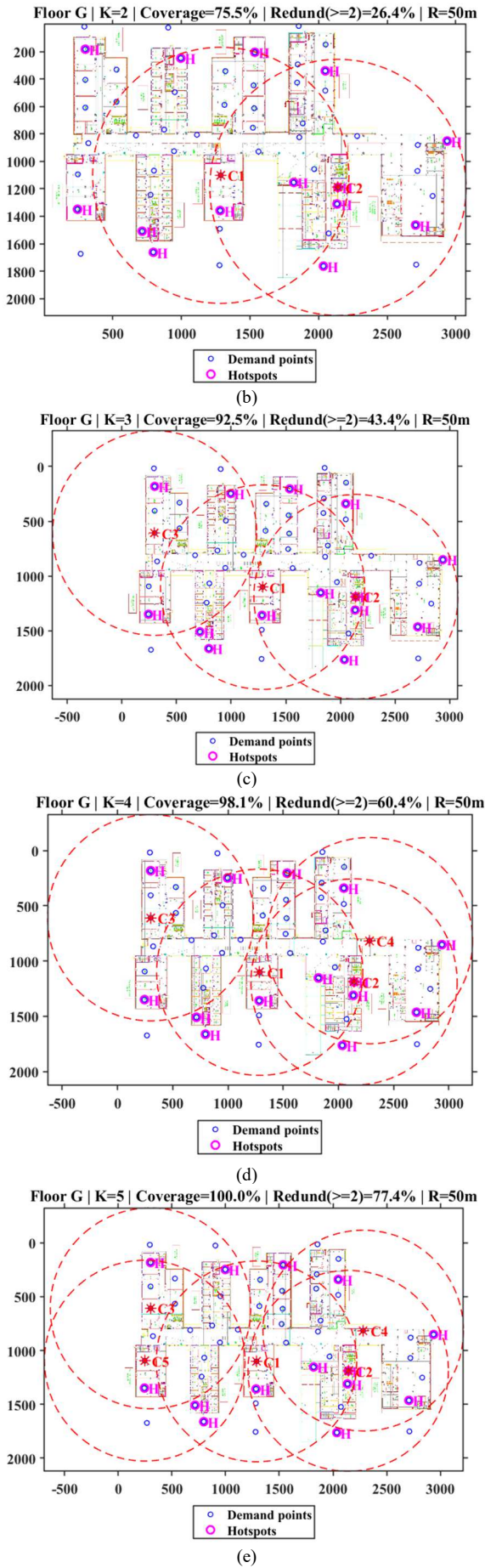


Fig. 4. Charger placement results for the KCST ground floor with $R=50$ m under different charger budgets (a) $K=1$ (b) $K=2$ (c) $K=3$ (d) $K=4$ (e) $K=5$

Table II presents the comparative results for the ground floor with $K=3$ and $R=50$ m. Greedy achieved identical coverage to exhaustive search while requiring only a fraction of the computation time, whereas random allocation yielded substantially poorer coverage. These findings confirm that greedy strikes the best balance between computational efficiency and solution quality, making it highly suitable for real-world planning at the campus scale.

Fig. 5 illustrates how coverage improves as the number of chargers on the Ground floor increases, given a fixed coverage radius of 50 m. The curve follows the expected diminishing returns pattern typical of submodular optimization problems.

Fig. 6 demonstrates how the achievable coverage varies with the number of chargers (K) when different coverage radii are assumed. R-sensitivity analysis highlights how charger reach directly impacts infrastructure needs. At the baseline ($R = 50$ m), three chargers provide over 92% coverage, with four achieving near-complete coverage. A smaller radius (40 m) requires more chargers and still fails to fully cover all points, while a larger radius (60 m) achieves full coverage with fewer chargers but assumes ideal, less practical conditions. Overall, the analysis confirms that while coverage scales predictably with additional chargers, the recommended number must be grounded in realistic assumptions about accessibility range.

On the first and second floors, fewer chargers are sufficient to achieve higher coverage percentages, as shown in Figs. 7 and 8. This is primarily because these floors do not include outdoor areas such as parking lots, event spaces, or sports facilities, which are present on the ground floor and increase the number of demand points.

TABLE II COMPARISON OF GREEDY, EXHAUSTIVE, AND RANDOM PLACEMENT METHODS (GROUND FLOOR, $K=3$, $R=50$ M).

Method	Coverage (%)	Redundancy (%)	Runtime (s)
Exhaustive Search	92.5	43.4	12.6
Greedy Algorithm	92.5	43.4	0.08
Random Placement	71.2 (avg)	15.6 (avg)	0.05

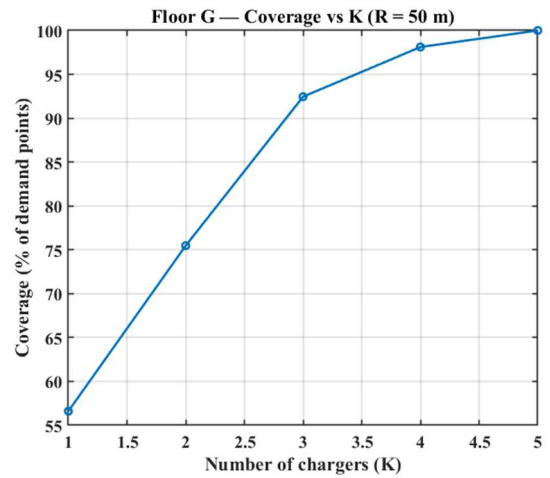


Fig. 5. Coverage performance on the KCST ground floor as a function of the number of chargers (K)

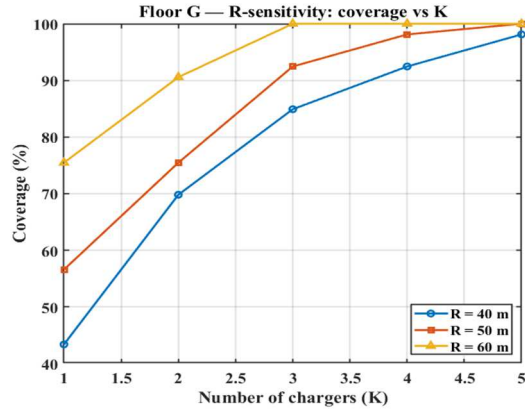


Fig. 6. R-sensitivity analysis of ground floor coverage as a function of the number of chargers (K)

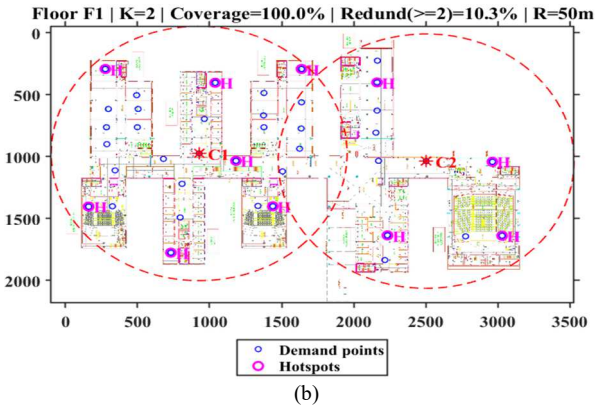
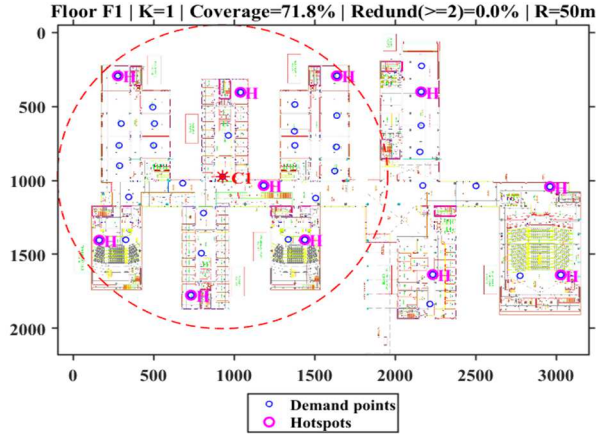


Fig. 7. Charger placement results for the KCST first floor with $R=50$ m under different charger budgets (a) $K=1$ (b) $K=2$

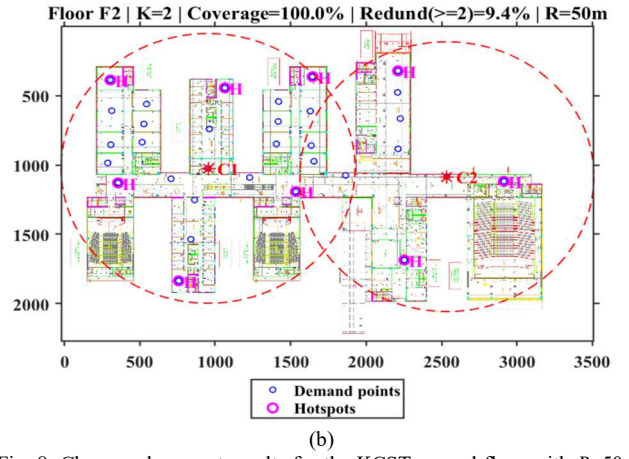
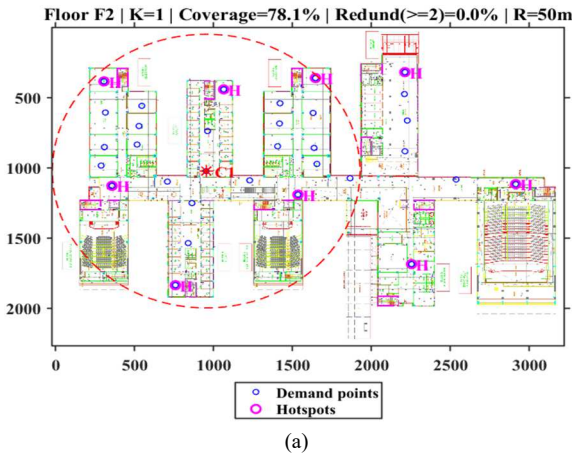


Fig. 8. Charger placement results for the KCST second floor with $R=50$ m under different charger budgets (a) $K=1$ (b) $K=2$

Beyond spatial coverage, deployment of wheelchair charging stations must also consider electrical load and user safety. Typical wheelchair chargers draw between 100–300 W per unit [16]. Thus, five simultaneously operating chargers impose a demand of approximately 0.5–1.5 kW. While modest relative to lighting or HVAC loads, this still necessitates careful distribution across campus panels to avoid local circuit overloading. International safety standards such as IEC 60335-2-29 (battery charger safety), ISO 7176-25 (wheelchair electrical requirements), and IEC 60364 (building electrical installations) recommend protective measures including residual current devices, short-circuit protection, and compliant socket height and accessibility [17–19]. Embedding these considerations ensures that the optimized placement strategy is not only spatially effective but also electrically robust and safe for everyday use.

These findings demonstrate that charger placement requirements are floor-dependent. The ground floor, due to its complexity and demand diversity, requires a larger number of chargers to balance both coverage and redundancy.

In contrast, the upper floors achieve full coverage with only two chargers, reflecting lower spatial complexity and concentrated demand (Table III).

TABLE III COVERAGE, REDUNDANCY, AND MARGINAL GAIN RESULTS FOR CHARGER PLACEMENT ACROSS THE CAMPUS

Floor / K (chargers)	Coverage (%)	Redundancy (%)	Marginal Gain (%)
Ground Floor			
K = 1	56.6	0.0	0.0
K = 2	75.5	26.4	18.9
K = 3	92.5	43.4	17.0
K = 4	98.1	60.4	5.6
K = 5	100	77.4	1.9
First Floor			
K = 1	71.8	0.0	0.0
K = 2	100	10.3	28.2
K = 3	100	10.3	0.0
K = 4	100	10.3	0.0
K = 5	100	10.3	0.0

Second Floor			
K = 1	78.1	0.0	0.0
K = 2	100	9.4	21.9
K = 3	100	9.4	0.0
K = 4	100	9.4	0.0
K = 5	100	9.4	0.0

From a planning perspective, this suggests a tiered infrastructure deployment strategy: allocate more chargers to high-demand, spatially complex floors (e.g., ground floor), while optimizing the number of installations on upper floors where coverage saturates quickly. Such a strategy balances efficiency, cost, and user accessibility, while ensuring resilience through redundancy on the busiest floor.

IV. CONCLUSION

This study presented a map-driven optimization framework for planning wheelchair charging stations on the university campus. By modeling charger allocation as a weighted maximal-coverage problem and solving it via a greedy heuristic, the method balances coverage, redundancy, and cost-efficiency while remaining simple and easy for implementation. The presented simulation results showed that three chargers suffice to cover over 92% of the ground floor, while just two chargers ensure complete coverage of the upper university floors. Sensitivity analyses confirmed the robustness of the framework and highlighted the impact of realistic coverage radius assumptions on infrastructure requirements. Beyond KCST, the framework offers a transferable and scalable solution for accessible energy infrastructure in educational, commercial, and public spaces. It enables decision-makers to make data-driven, inclusive, and cost-conscious planning choices, reinforcing commitments to universal design and sustainable campus development.

As future work, the framework can be extended to integrate renewable energy sources such as PV-powered charging stations and smart scheduling mechanisms, enhancing both sustainability and operational efficiency. Furthermore, incorporating dynamic demand patterns driven by class schedules, events, or occupancy data could refine placement strategies beyond static maps, ensuring that accessibility is matched to real usage rhythms in campus life.

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